AZURE MEETUP PUGLIA

Santa Cloud Day

Lecce, 20 Dicembre 2025

Università del Salento,
Polo Urbano Studium 2000

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Devcontainers: from Local to Cloud

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What’s the story?

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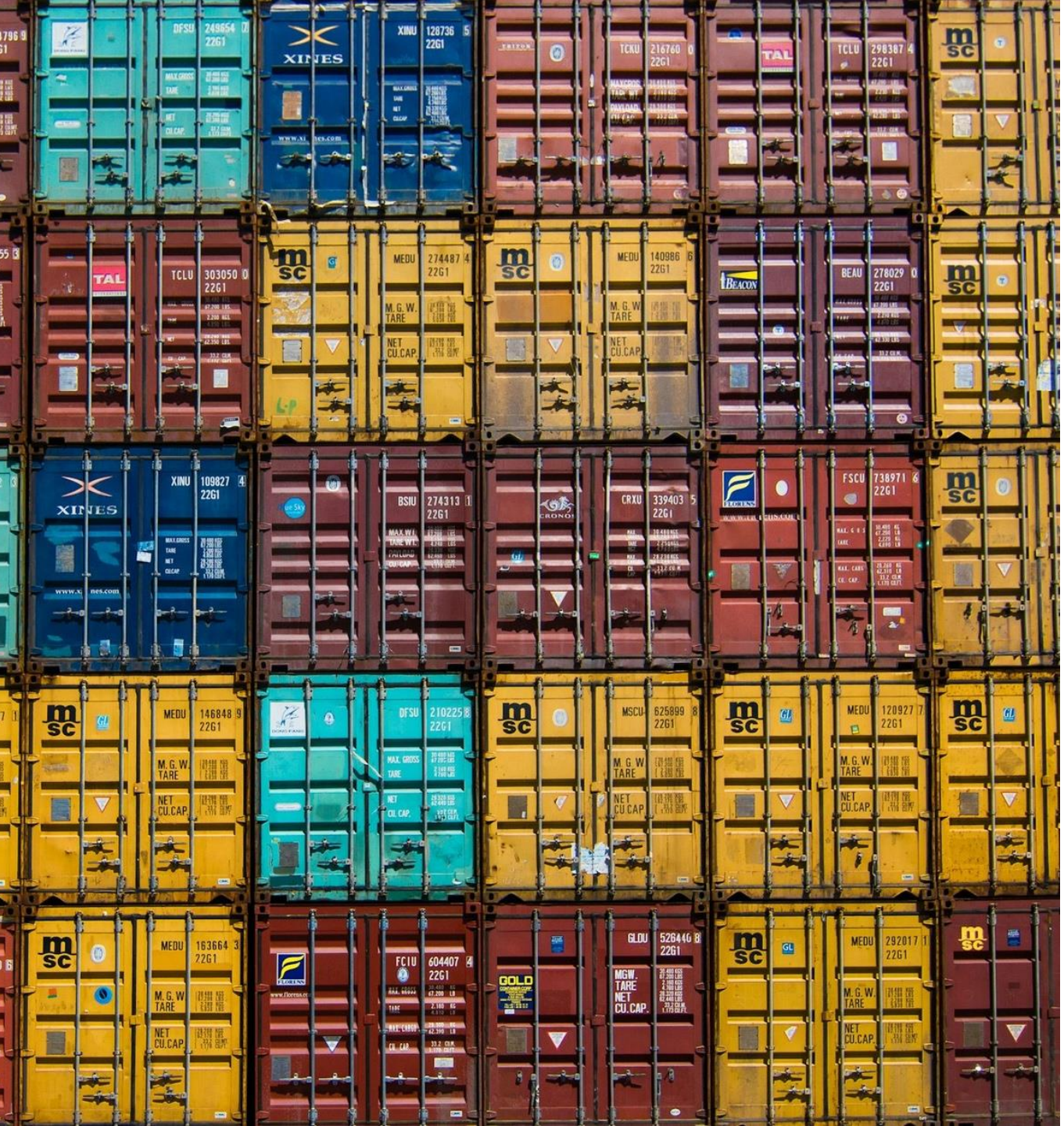


Creating replicable development environments

(Spoiler in the pic)

The main issue

QUESTION.

“

How many projects have you been working on in recent months?

A lot of projects

My personal answer

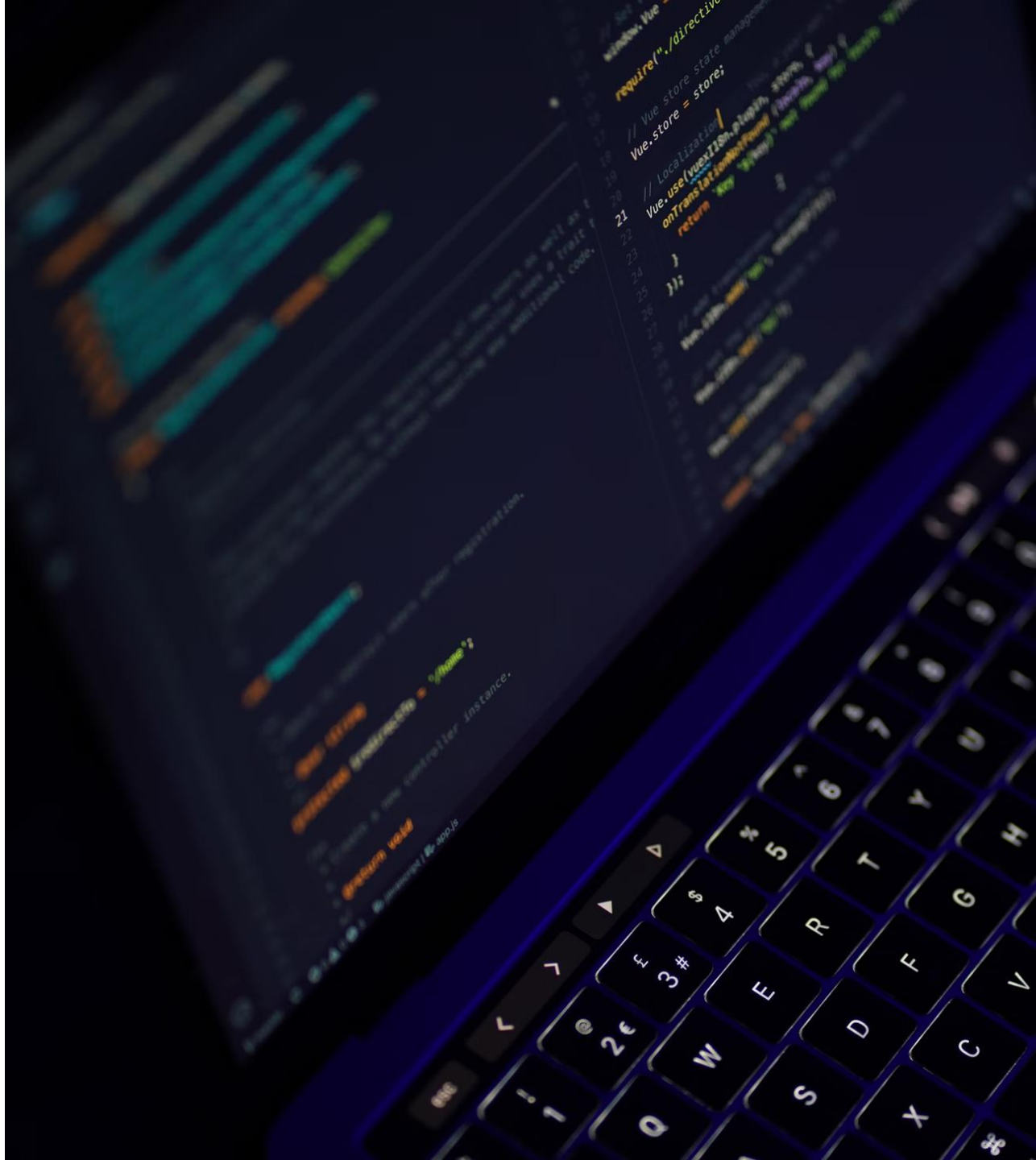
“

How many setups have you done in the last few months?

Too many setups

My personal answer too

Project setup



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Project setup

- Enviroment/Runtime
- Lib with a specific version
- Buildchain
- Local registry
- IDE Configuration

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Project setup

- Complex process
- Time consuming
- Error prone
- Non-automated process
- Inconsistency

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Solution

Would Docker solve this problem?

Absolutely (but...)

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VSCode Remote Development

Developing within a remote environment

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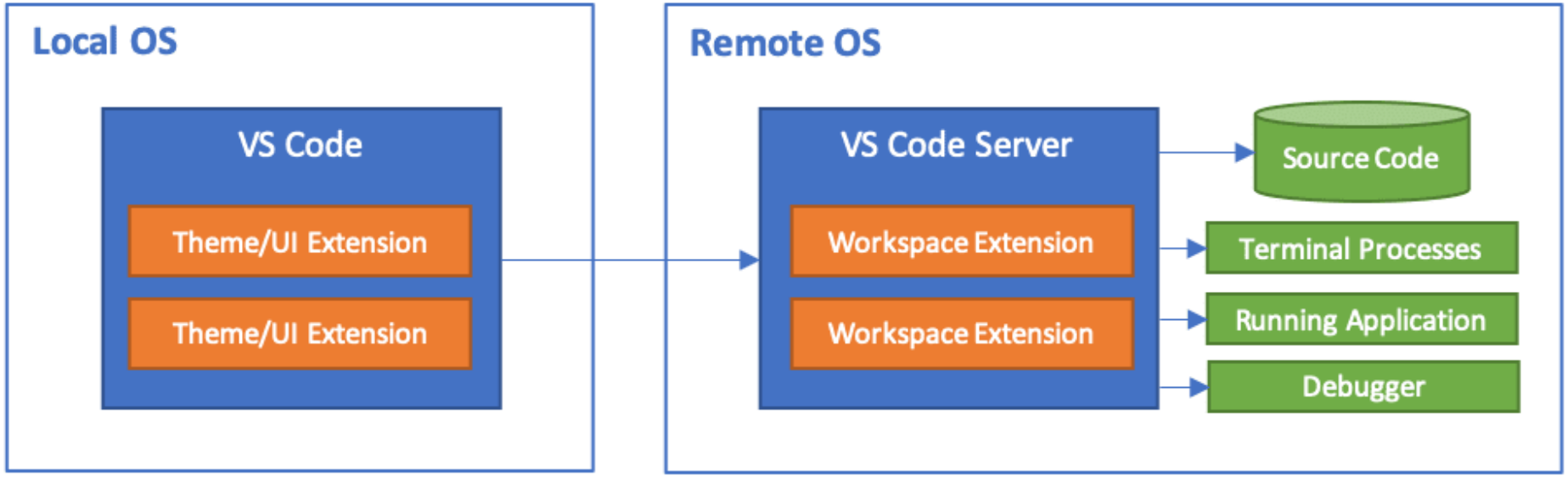
Visual Studio Code Remote Development allows you to use a container, remote machine, or the Windows Subsystem for Linux (WSL) as a full-featured development environment.



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VS Code Remote Development

- Developing in a production-like environment
- Segregated development environment
- Faster onboarding
- Easier runtime management
- Development in Linux environment on Windows via WSL

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Devcontainers

A container-based solution

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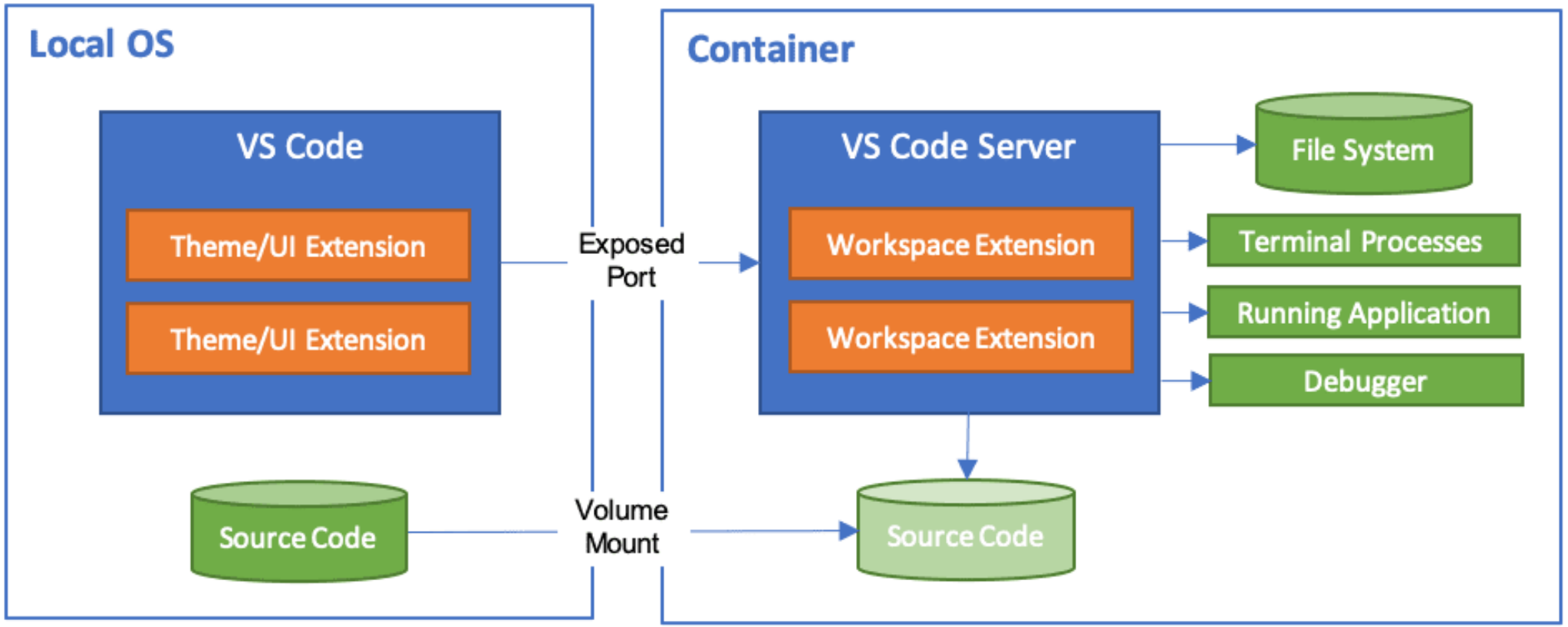

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Devcontainers

Install Dev Containers extension

Simply install a VSCode extension that enables it to run Devcontainers

Define your devcontainer.json

Define your environment, your runtimes and your IDE configuration

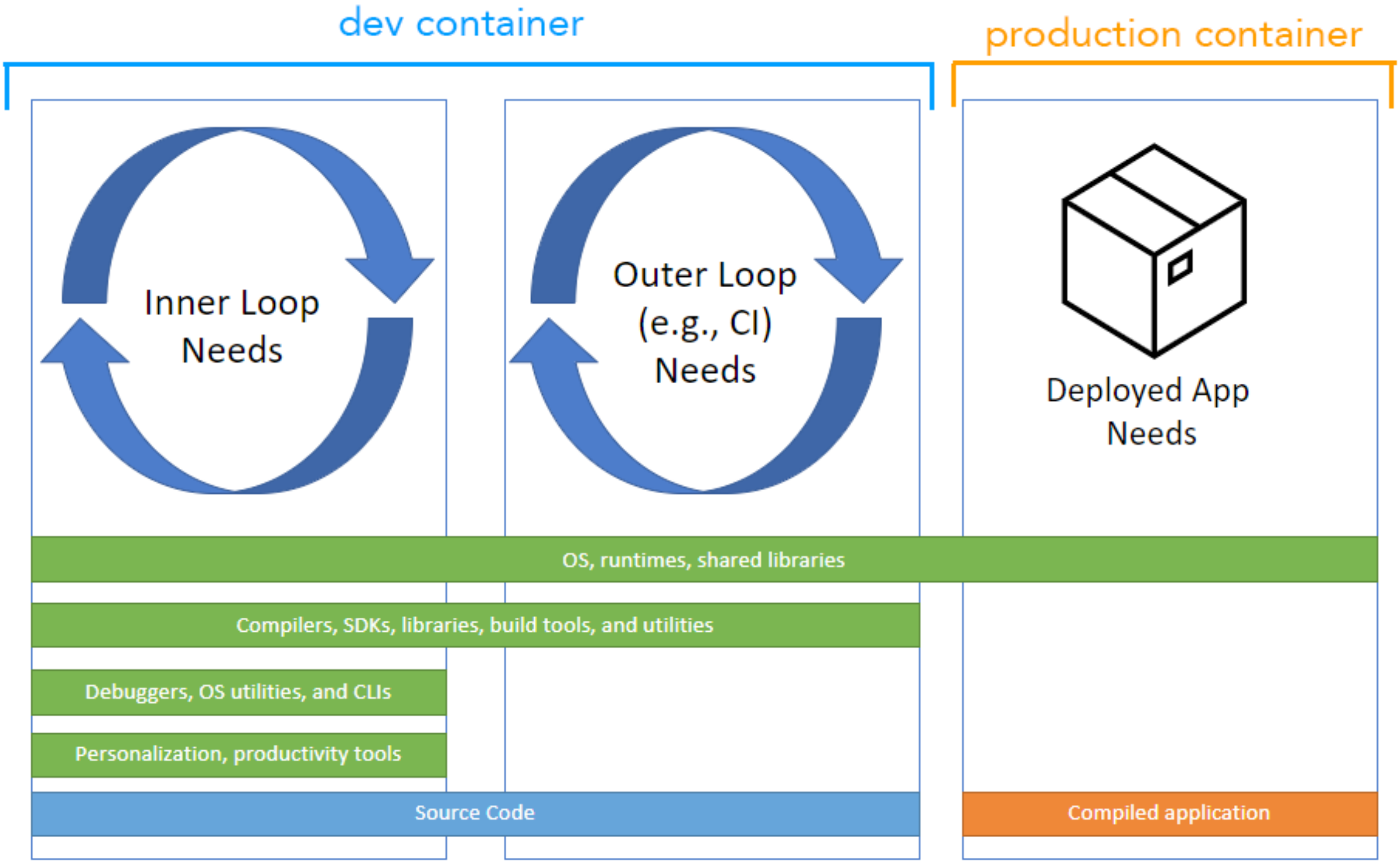
Open VSCode in Container

Build a docker image and run the container. Attach VSCode to the created container.

Devcontainers

devcontainer.json

- Environment setup Docker-based
- Automation
- Hooks
- IDE setup


dev container
production container
Inner Loop
Needs
Outer Loop
(e.g., CI)
Needs
Deployed App
Needs
OS, runtimes, shared libraries
Compilers, SDKs, libraries, build tools, and utilities
Debuggers, OS utilities, and CLIs
Personalization, productivity tools
Source Code
Compiled application

Is Devcontainers strictly tied with VS Code?

I know, someone is wondering this question

Devcontainer as standard de-facto

Devcontainers / Editors

- VSCode
- Visual Studio (> 2022)
- IntelliJ IDEA

Devcontainers / Tools

- CLI
- Cachix devenv
- Jetify (*nix based)

Devcontainers / Services

- Github Codespaces
- CodeSandbox
- Devpod

Demo time

Wrap-Up

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Wrap Up

PROs (1/2)

- Repeatable setup process
- Automated environment configuration
- Isolation
- Self-consistency
- Runtimes/libs management
- Automated IDE setup

Wrap Up

PROs (2/2)

- Standard de-facto
- CLI
- WSL (?)

Wrap Up

CONs

- Complex setups
- Weak performance with WSL
- Hard-disk space eater
- Cryptic error messages

Q&A

Grazie

https://meetme.cards/p/valeriocomo

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